

Generation of Jacobi Matrices on the Simplex

d=1

```
In[1]:= ClearAll[P,W,Wab,Wabc,H,a,b,c,h,Ax,Bx,Ay,By,A1,A2,B1,B2,x,y,Jx,Jy,Antmp];
13
14 (* Orthogonal Jacobi polynomials in 1D *)
15 P[a_,b_,n_,x_] :=
16 Assuming[
17
18     Element[n, Integers] &&
19     Element[a, Reals] &&
20     Element[b, Reals] &&
21     Element[x, Reals] &&
22     n ≥ 0 &&
23     a > -1 &&
24     b > -1 &&
25     x ≥ -1 &&
26     x ≤ 1,
27
28     ((-1)^n)/(2^n * n!) *
29     (1-x)^(-a)*(1+x)^(-b) *
30     D[(1-x)^(a+n)*(1+x)^(b+n),{x,n}]
31
32 ];
33
34 (* Pi,Pj are orthogonal in L2([-1,1],W) weighted inner product *)
35 W[a_,b_,x_] :=
36 Assuming[
37
38     Element[a, Reals] &&
39     Element[b, Reals] &&
40     Element[x, Reals] &&
41     a > -1 &&
42     b > -1 &&
43     x ≥ -1 &&
```

```

      x ≤ 1,

      ((1-x)^a) * ((1+x)^b)

];

(* <1,1> = 1 only if we divide inner products by Wab *)
Wab[a_,b_] :=
Assuming[

  Element[a, Reals] &&
  Element[b, Reals] &&
13   a > -1 &&
14   b > -1,
15
16   Integrate[W[a,b,x],{x,-1,1}]
17
18 ];
19
20 (* Use H to check orthogonality, and
21 compute structural constants for Pk *)
22 H[a_,b_,i_,j_] :=
23 Assuming[
24
25   Element[a, Reals] &&
26   Element[b, Reals] &&
27   Element[i, Integers] &&
28   Element[j, Integers] &&
29   a > -1 &&
30   b > -1 &&
31   i ≥ 0 &&
32   j ≥ 0,
33
34   Integrate[P[a,b,i,x]*P[a,b,j,x]*W[a,b,x], {x,-1,1}]/Wab[a,b]
35
36 ];
37
38 (* normalize the polynomials *)
39 p[a_,b_,n_,x_] :=
40 Assuming[
41
42   Element[a, Reals] &&
43   Element[b, Reals] &&
44

```

```

        Element[n, Integers] &&
        Element[x, Reals] &&
        a > -1 &&
        b > -1 &&
        n ≥ 0 &&
        x ≥ -1 &&
        x ≤ 1,

        P[a,b,n,x] / Sqrt[H[a,b,n,n]]
    ];

```

13

14

```

106 (* check if we've done everything right so far *)
107 a = 1; b = 3;
108 h[i_,j_] := Integrate[p[a,b,i,x]*p[a,b,j,x]*W[a,b,x],{x,-1,1}]/Wab[a,b];
109 (* expect identity *)
110 MatrixForm[Table[h[i,j],{i,0,5},{j,0,5}]]

```

d = 2

```

121 (* now move on to Tri *)
122 ClearAll[a,b,c,x,y,n,k,Sz]
123 (* Normalization for the measure *)
124 Wabc[a_,b_,c_] :=
125 Assuming[
126
127     Element[a, Reals] &&
128     Element[b, Reals] &&
129     Element[c, Reals] &&
130     a > -1/2 &&
131     b > -1/2 &&
132     c > -1/2,
133
134     Gamma[a+b+c+3/2]/(Gamma[a+1/2]*Gamma[b+1/2]*Gamma[c+1/2])
135
136 ];
137 (* Normalized measure *)
138 W[a_,b_,c_,x_,y_] :=

```

```

Assuming[

  Element[a, Reals] &&
  Element[b, Reals] &&
  Element[c, Reals] &&
  Element[x, Reals] &&
  Element[y, Reals] &&
  a > -1/2 &&
  b > -1/2 &&
  c > -1/2 &&
  x ≥ 0 &&
  y ≥ 0 &&
  1-x-y ≥ 0,

  x^(a-1/2)*y^(b-1/2)*(1-x-y)^(c-1/2)*Wabc[a,b,c]

];

(* structure constants *)

H[a_,b_,c_,k_,n_]:=
Assuming[

  Element[a, Reals] &&
  Element[b, Reals] &&
121  Element[c, Reals] &&
122  Element[k, Integers] &&
123  Element[n, Integers] &&
124  a > -1/2 &&
125  b > -1/2 &&
126  c > -1/2 &&
127  k ≥ 0 &&
128  n ≥ 0,
129
130  Piecewise[
131    {{Sqrt[Wabc[a,b,c]/((2*n+a+b+c+1/2)*(2*k+b+c)) *
132    Gamma[n+k+b+c+1]*Gamma[n-k+a+1/2]*Gamma[k+b+1/2]*Gamma[k+c+1/2] /
133    ((n-k)!*k!*Gamma[n+k+a+b+c+1/2]*Gamma[k+b+c])], k ≤ n},{0,k>n}}]
134
135  ];
136
137
138  (* Orthonormal Koornwinder polynomials *)

```

```

P[a_,b_,c_,k_,n_,x_,y_]:=
Assuming[

    Element[a, Reals] &&
    Element[b, Reals] &&
    Element[c, Reals] &&
    Element[k, Integers] &&
    Element[n, Integers] &&
    Element[x, Reals] &&
    Element[y, Reals] &&
    a > -1/2 &&
    b > -1/2 &&
    c > -1/2 &&
    k ≥ 0 &&
    n ≥ 0 &&
    x ≥ 0 &&
    y ≥ 0 &&
    1-x-y ≥ 0,

    H[a,b,c,k,n]^(-1) *
    (P[2*k+b+c,a-1/2,n-k,z]/.z→2*x-1) *
    (1-x)^k *
    (P[c-1/2,b-1/2,k,z]/.z→2*y/(1-x)-1)

];
121
122
123
124
...

220 (* check if we've done everything right so far *)
221
222 (* expect upper trimat*)
223 MatrixForm[Table[H[1/2,1/2,1/2,k,n],{k,0,10},{n,0,10}]]
224 a = 1/2; b = 1/2; c = 1/2 ;
225 h[i_,j_] := Integrate[P[a,b,c,i,i,x,y]*P[a,b,c,j,j,x,y]*W[a,b,c,x,y],{x,0,1},{y,0,1-x}];
226 (* expect identity *)
227 MatrixForm[Table[h[i,j],{i,0,5},{j,0,5}]]

230 (* our lex order is (x^n,x^{n-1}y,...,y^n) *)
231 ClearAll[a,b,c,x,y,n,nn,k,Sz,Jn1,Jn2,B1,B2,inds,inds1,An1,An2,Bn1,Bn2,A1,A2]

```

```
(* Vectorization for space of homogeneous polynomials of degree n *)
```

```
P[a_,b_,c_,n_,x_,y_] :=
```

```
Assuming[
```

```
Element[a, Reals] &&
```

```
Element[b, Reals] &&
```

```
Element[c, Reals] &&
```

```
Element[n, Integers] &&
```

```
Element[x, Reals] &&
```

```
Element[y, Reals] &&
```

```
a > -1/2 &&
```

```
b > -1/2 &&
```

```
c > -1/2 &&
```

```
n ≥ 0 &&
```

```
x ≥ 0 &&
```

```
y ≥ 0 &&
```

```
1-x-y ≥ 0,
```

```
Table[P[a,b,c,k,n,x,y],{k,0,n}]
```

```
]
```

```
(* Blocks of Jacobi matrix *)
```

```
Ax[a_,b_,c_,n_] :=
```

```
Assuming[
```

```
Element[a, Reals] &&
```

```
Element[b, Reals] &&
```

```
Element[c, Reals] &&
```

```
Element[n, Integers] &&
```

```
a > -1/2 &&
```

```
b > -1/2 &&
```

```
c > -1/2 &&
```

```
n ≥ 0,
```

```
Table[Integrate[Outer[Times,
```

```
x*P[a,b,c,k,x,y],
```

```
P[a,b,c,k+1,x,y]]*W[a,b,c,x,y],
```

```
{x,0,1},{y,0,1-x},{k,0,n-2}]
```

```
];
```

```
Ay[a_,b_,c_,n_] :=
Assuming[

  Element[a, Reals] &&
  Element[b, Reals] &&
  Element[c, Reals] &&
  Element[n, Integers] &&
  a > -1/2 &&
  b > -1/2 &&
  c > -1/2 &&
  n ≥ 0,

  Table[Integrate[Outer[Times,
y*P[a,b,c,k,x,y],
P[a,b,c,k+1,x,y]]*W[a,b,c,x,y],
{x,0,1},{y,0,1-x}},{k,0,n-2}]

];
```

```
Bx[a_,b_,c_,n_] :=
Assuming[

  Element[a, Reals] &&
  Element[b, Reals] &&
  Element[c, Reals] &&
  Element[n, Integers] &&
  a > -1/2 &&
  b > -1/2 &&
  c > -1/2 &&
  n ≥ 0,

  Table[Integrate[Outer[Times,
x*P[a,b,c,k,x,y],
P[a,b,c,k,x,y]]*W[a,b,c,x,y],
{x,0,1},{y,0,1-x}},{k,0,n-1}]

];
```

```
By[a_,b_,c_,n_] :=
Assuming[
```

```
230   Element[a, Reals] &&
231   Element[b, Reals] &&
```

```

Element[c, Reals] &&
Element[n, Integers] &&
a > -1/2 &&
b > -1/2 &&
c > -1/2 &&
n ≥ 0,

Table[Integrate[Outer[Times,
y*P[a,b,c,k,x,y],
P[a,b,c,k,x,y]]*W[a,b,c,x,y],
{x,0,1},{y,0,1-x},{k,0,n-1}]
];

```

```

335 (* generate block recurrence matrices *)
336 a = 1/2; b = 1/2; c = 1/2; n = 2;
337 B1 = Bx[a,b,c,n];
338 A1 = Ax[a,b,c,n];
339 B2 = By[a,b,c,n];
340 A2 = Ay[a,b,c,n];
341

```

```

344 B1//MatrixForm

```

```

347 (* assemble into Jacobi matrices for each dim *)
348 n = 3;
349 Sz = n *(n+1)/2;
350 Jn1 = ConstantArray[0,{Sz,Sz}];
351 Jn2 = ConstantArray[0,{Sz,Sz}];
352 inds = {1,1};
353 For [nn = 1, nn ≤ n-1, nn++,
354   Jn1[[inds[[1]];inds[[2]],inds[[1]];inds[[2]]] = B1[[nn]];
355   Jn2[[inds[[1]];inds[[2]],inds[[1]];inds[[2]]] = B2[[nn]];
356   inds1 = inds + {nn,nn+1};
357   Jn1[[inds[[1]];inds[[2]],inds1[[1]];inds1[[2]]] = A1[[nn]];
358   Jn1[[inds1[[1]];inds1[[2]],inds[[1]];inds[[2]]] = Transpose[A1[[nn]]];
359   Jn2[[inds[[1]];inds[[2]],inds1[[1]];inds1[[2]]] = A2[[nn]];
360   Jn2[[inds1[[1]];inds1[[2]],inds[[1]];inds[[2]]] = Transpose[A2[[nn]]];
361   inds = inds + {nn,nn+1}
362 ];
363 Jn1[[inds[[1]];inds[[2]],inds[[1]];inds[[2]]] = B1[[n]];
364 Jn2[[inds[[1]];inds[[2]],inds[[1]];inds[[2]]] = B2[[n]];

```



```
MatrixPlot[Jn1]
MatrixPlot[Jn2]
```

$d = 3$, the tetrahedron awaits!

```
392 (* Now we define orthogonal polynomials on the tetrahedron
393
394 We proceed analogously to 2D in that we only need to multiply
395 the 2D polynomials by a specific form of the 1D polynomials
396 to generate the 3D polynomials. However, we just unroll this
397 all in terms of the 1D polynomials.
398 *)

In[7]:= (* un-normalized measure *)
402 W[a_,b_,c_,d_,x_,y_,z_]:=
403 Assuming[
404
405     Element[a, Reals] &&
406     Element[b, Reals] &&
407     Element[c, Reals] &&
408     Element[d, Reals] &&
409     Element[x, Reals] &&
410     Element[y, Reals] &&
411     Element[z, Reals] &&
412     a > -1/2 &&
413     b > -1/2 &&
414     c > -1/2 &&
415     d > -1/2 &&
416     x ≥ 0 &&
417     y ≥ 0 &&
418     z ≥ 0 &&
419     1-x-y-z ≥ 0,
420
421     x^(a-1/2)*y^(b-1/2)*z^(c-1/2)*(1-x-y-z)^(d-1/2)
422
423 ];
424
```

```
(* Normalization for the measure *)
Wabcd[a_,b_,c_,d_] :=
Assuming[

    Element[a, Reals] &&
    Element[b, Reals] &&
    Element[c, Reals] &&
    Element[d, Reals] &&
    a > -1/2 &&
    b > -1/2 &&
    c > -1/2 &&
    d > -1/2,

    Integrate[W[a,b,c,d,x,y,z],{x,0,1},{y,0,1-x},{z,0,1-x-y}]
];
```

```
(* normalized measure *)
w[a_,b_,c_,d_,x_,y_,z_] :=
Assuming[
```

```
402
403     Element[a, Reals] &&
404     Element[b, Reals] &&
405     Element[c, Reals] &&
406     Element[d, Reals] &&
407     Element[x, Reals] &&
408     Element[y, Reals] &&
409     Element[z, Reals] &&
410     a > -1/2 &&
411     b > -1/2 &&
412     c > -1/2 &&
413     d > -1/2 &&
414     x ≥ 0 &&
415     y ≥ 0 &&
416     z ≥ 0 &&
417     1-x-y-z ≥ 0,
418
419     x^(a-1/2)*y^(b-1/2)*z^(c-1/2)*(1-x-y-z)^(d-1/2)/Wabcd[a,b,c,d]
420
421 ];
422
423 a = 1/2; b = 1/2; c = 1/2; d = 1/2;
424
425
```

```

In[11]:= (* structure constants *)
471
472 H[a_,b_,c_,d_,k_,j_,n_] :=
473 Assuming[
474
475     Element[a, Reals] &&
476     Element[b, Reals] &&
477     Element[c, Reals] &&
478     Element[k, Integers] &&
479     Element[n, Integers] &&
480     a > -1/2 &&
481     b > -1/2 &&
482     c > -1/2 &&
483     d > -1/2 &&
484     k ≥ 0 &&
485     j ≥ 0 &&
486     n ≥ 0,
487
488     alphaj = {n-k,k-j,j};
489     kj = {a,b,c,d};
490     kjl1 = {a+b+c+d,b+c+d,c+d,d};
491     alphj1 = {n,k,j,0};
492     aj = {2*k+b+c+d+1/2,2*j+c+d,d-1/2};
493     Sqrt[Product[
494         Pochhammer[kj[[1]]+1/2,alphaj[[1]]*
495         Pochhammer[kjl1[[1]]+(1/2)*(3-l+1),alphj1[[1]]+alphj1[[1+1]]/
496         (alphaj[[1]]*Pochhammer[kjl1[[1]]+(1/2)*(3-l+2),alphj1[[1]]+alphj1[[1+1]])*
497         (2*(aj[[1]]+kj[[1]]+alphaj[[1]]+1)/
498         (2*(aj[[1]]+kj[[1]]+2*alphaj[[1]]+1),
499         {l,1,3}]]
500
501 ];
502

```

```

In[12]:= (* Orthonormal Jacobi polynomials on the tetrahedron *)
506
507 P[a_,b_,c_,d_,n_,k_,j_,x_,y_,z_]:=
508 Assuming[
509
510     Element[a, Reals] &&
511     Element[b, Reals] &&
512     Element[c, Reals] &&
513     Element[d, Reals] &&
514     Element[n, Integers] &&
515     Element[k, Integers] &&
516     Element[j, Integers] &&
517     Element[x, Reals] &&
518     Element[y, Reals] &&
519     Element[z, Reals] &&
520     a > -1/2 &&
521     b > -1/2 &&
522     c > -1/2 &&
523     d > -1/2 &&
524     k ≥ 0 &&
525     n ≥ 0 &&
526     x ≥ 0 &&
527     y ≥ 0 &&
528     z ≥ 0 &&
529     1-x-y-z ≥ 0,
530
531     H[a,b,c,d,k,j,n]^(-1) *
532     (P[2*k+b+c+d+1/2,a-1/2,n-k,q]/.q→2x-1) *
533     (1-x)^(k-j)*(P[2*j+c+d,b-1/2,k-j,q]/.q→(2*y/(1-x)-1)) *
534     (1-x-y)^j*(P[d-1/2,c-1/2,j,q]/.q→(2*z/(1-x-y)-1))
535 ];

538 (* check that we've done everything right so far *)
539 htet[i_,j_] := Integrate[P[a,b,c,d,i,0,0,x,y,z]*P[a,b,c,d,j,0,0,x,y,z]*w[a,b,c,d,x,y,z],{x,0,
540 (* expect identity *)
541 MatrixForm[Table[htet[i,j],{i,0,5},{j,0,5}]]
542

In[13]:= (* Vectorization for space of homogeneous polynomials of degree n *)
546 P[a_,b_,c_,d_,n_,x_,y_,z_]:=
547 Assuming[
548
549     Element[a, Reals] &&

```

```

Element[b, Reals] &&
Element[c, Reals] &&
Element[n, Integers] &&
Element[x, Reals] &&
Element[y, Reals] &&
a > -1/2 &&
b > -1/2 &&
c > -1/2 &&
d > -1/2 &&
n ≥ 0 &&
x ≥ 0 &&
y ≥ 0 &&
z ≥ 0 &&
1-x-y-z ≥ 0,

```

```

ArrayFlatten[ArrayFlatten[Table[P[a,b,c,d,n,k,j,x,y,z],{k,0,n},{j,0,k}],1],1]

```

```

];

```

```

(* Blocks of Jacobi matrix *)

```

```

Ax[a_,b_,c_,d_,n_] :=

```

```

Assuming[

```

```

Element[a, Reals] &&
Element[b, Reals] &&
Element[c, Reals] &&
Element[d, Reals] &&
Element[n, Integers] &&
a > -1/2 &&
b > -1/2 &&
c > -1/2 &&
d > -1/2 &&
n ≥ 0,

```

```

Table[Integrate[Outer[Times,
x*P[a,b,c,d,k,x,y,z],
P[a,b,c,d,k+1,x,y,z]]*W[a,b,c,d,x,y,z],
{x,0,1},{y,0,1-x},{z,0,1-x-y}],{k,0,n-2}]

```

```

];

```

```

Ay[a_,b_,c_,d_,n_] :=

```

```

Assuming[

```

546

547

548

549

```

Element[a, Reals] &&
Element[b, Reals] &&
Element[c, Reals] &&
Element[d, Reals] &&
Element[n, Integers] &&
a > -1/2 &&
b > -1/2 &&
c > -1/2 &&
d > -1/2 &&
n ≥ 0,

Table[Integrate[Outer[Times,
y*P[a,b,c,d,k,x,y,z],
P[a,b,c,d,k+1,x,y,z]]*W[a,b,c,d,x,y,z],
{x,0,1},{y,0,1-x},{z,0,1-x-y}},{k,0,n-2}

];

```

```

Az[a_,b_,c_,d_,n_] :=
Assuming[

```

```

Element[a, Reals] &&
Element[b, Reals] &&
Element[c, Reals] &&
Element[d, Reals] &&
Element[n, Integers] &&
a > -1/2 &&
b > -1/2 &&
c > -1/2 &&
d > -1/2 &&
n ≥ 0,

Table[Integrate[Outer[Times,
z*P[a,b,c,d,k,x,y,z],
P[a,b,c,d,k+1,x,y,z]]*W[a,b,c,d,x,y,z],
{x,0,1},{y,0,1-x},{z,0,1-x-y}},{k,0,n-2}

];

```

546

547

```
Bx[a_,b_,c_,d_,n_] :=
```

548

```
Assuming[
```

549

```

Element[a, Reals] &&
Element[b, Reals] &&
Element[c, Reals] &&
Element[d, Reals] &&
Element[n, Integers] &&
a > -1/2 &&
b > -1/2 &&
c > -1/2 &&
d > -1/2 &&
n ≥ 0,

Table[Integrate[Outer[Times,
x*P[a,b,c,d,k,x,y,z],
P[a,b,c,d,k,x,y,z]]*W[a,b,c,d,x,y,z],
{x,0,1},{y,0,1-x},{z,0,1-x-y}},{k,0,n-1}

];

```

```

By[a_,b_,c_,d_,n_] :=
Assuming[

Element[a, Reals] &&
Element[b, Reals] &&
Element[c, Reals] &&
Element[d, Reals] &&
Element[n, Integers] &&
a > -1/2 &&
b > -1/2 &&
c > -1/2 &&
d > -1/2 &&
n ≥ 0,

Table[Integrate[Outer[Times,
y*P[a,b,c,d,k,x,y,z],
P[a,b,c,d,k,x,y,z]]*W[a,b,c,d,x,y,z],
{x,0,1},{y,0,1-x},{z,0,1-x-y}},{k,0,n-1}

];

```

```
546 Bz[a_,b_,c_,d_,n_] :=
```

```
547 Assuming[
```

```
548
```

```
549 Element[a, Reals] &&
```

```

Element[b, Reals] &&
Element[c, Reals] &&
Element[d, Reals] &&
Element[n, Integers] &&
a > -1/2 &&
b > -1/2 &&
c > -1/2 &&
d > -1/2 &&
n ≥ 0,

Table[Integrate[Outer[Times,
z*P[a,b,c,d,k,x,y,z],
P[a,b,c,d,k,x,y,z]]*W[a,b,c,d,x,y,z],
{x,0,1},{y,0,1-x},{z,0,1-x-y}},{k,0,n-1}

];

```

```

In[20]:= n = 5;
699 A1 = Ax[a,b,c,d,n];
700 B1 = Bx[a,b,c,d,n];
701 B2 = By[a,b,c,d,n];
702 A2 = Ay[a,b,c,d,n];
703 B3 = Bz[a,b,c,d,n];
704 A3 = Az[a,b,c,d,n];
705

```



```

In[27]:= (* assemble into Jacobi matrices for each dim *)
709 n = 5; dim=3;
710 Sz = Binomial[n-1+dim,n-1];
711 Jn1 = ConstantArray[0,{Sz,Sz}];
712 Jn2 = ConstantArray[0,{Sz,Sz}];
713 Jn3 = ConstantArray[0,{Sz,Sz}];
714 inds = {1,1}; bsz = 1;
715 For [nn = 1, nn ≤ n-1, nn++,
716     Jn1[[inds[[1]];inds[[2]],inds[[1]];inds[[2]]] = B1[[nn]];
717     Jn2[[inds[[1]];inds[[2]],inds[[1]];inds[[2]]] = B2[[nn]];
718     Jn3[[inds[[1]];inds[[2]],inds[[1]];inds[[2]]] = B3[[nn]];
719     bsz = bsz + nn + 1;
720     inds1 = inds + {bsz-(nn+1),bsz};
721     Jn1[[inds1[[1]];inds1[[2]],inds1[[1]];inds1[[2]]] = A1[[nn]];
722     Jn1[[inds1[[1]];inds1[[2]],inds1[[1]];inds1[[2]]] = Transpose[A1[[nn]];
723     Jn2[[inds1[[1]];inds1[[2]],inds1[[1]];inds1[[2]]] = A2[[nn]];
724     Jn2[[inds1[[1]];inds1[[2]],inds1[[1]];inds1[[2]]] = Transpose[A2[[nn]];
725     Jn3[[inds1[[1]];inds1[[2]],inds1[[1]];inds1[[2]]] = A3[[nn]];
726     Jn3[[inds1[[1]];inds1[[2]],inds1[[1]];inds1[[2]]] = Transpose[A3[[nn]];
727     inds = inds + {bsz-(nn+1),bsz};
728 ];
729 Jn1[[inds[[1]];inds[[2]],inds[[1]];inds[[2]]] = B1[[n]];
730 Jn2[[inds[[1]];inds[[2]],inds[[1]];inds[[2]]] = B2[[n]];
731 Jn3[[inds[[1]];inds[[2]],inds[[1]];inds[[2]]] = B3[[n]];
732

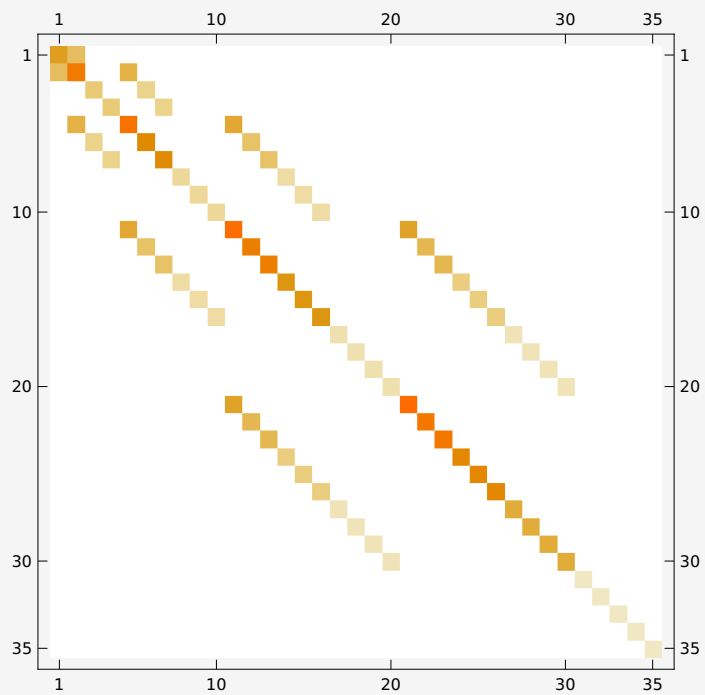
```

```

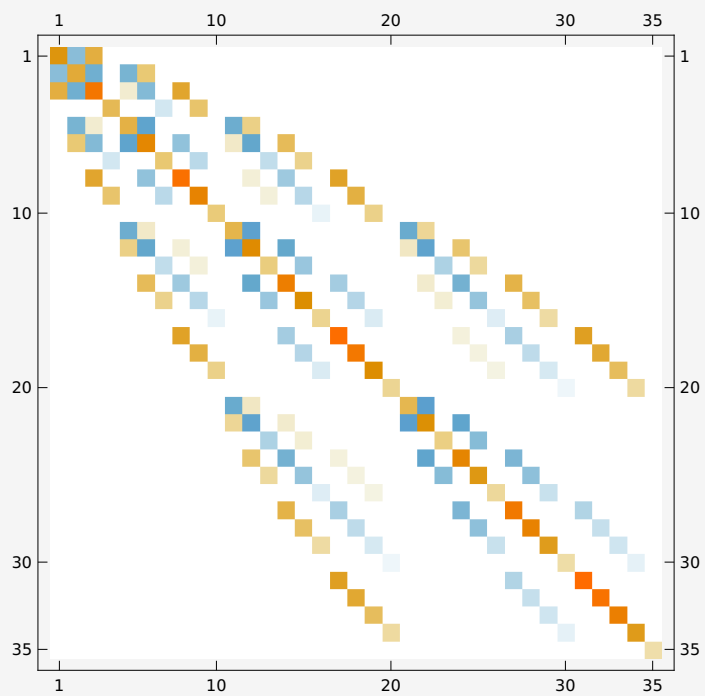
In[37]:= MatrixPlot[Jn1]
736 MatrixPlot[Jn2]
737 MatrixPlot[Jn3]
738

```

Out[37]=



Out[38]=



Out[39]=

